

Python Assignment

Question 1 — Star Pattern

Write a Python program that takes an integer n from the user and prints the following star pattern using nested loops.

For $n = 5$, the output should be:

```
*  
**  
***  
****  
*****
```

Requirement: The program should work for any positive value of n . Do not hard-code the pattern.

Question 2 — Matrix Operations

Write a Python program to input a 3×3 matrix from the user and perform the following operations:

1. Display the matrix in proper row-and-column form.
2. Calculate and display the sum of all elements.
3. Calculate and display the sum of the main diagonal elements.
4. Find and display the largest and smallest element.
5. Display the transpose of the matrix.

Example input:

```
1 2 3  
4 5 6  
7 8 9
```

Requirement: Use Python lists and loops. Do not use NumPy.